Machine Learning for Finance:

Auditing the AI Mirage, Regularization & Risk

Feature Engineering, Leakage Audits, and Holdout Showdowns Across 5 Global Assets

Georgios Kompothanasis

FinTech & Financial Econometrics
Executive Master's Presentation

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Asset Universe: SPY (US Equities) · EUR/USD (FX) · Bitcoin (BTC) · ATHEX (Greek Equities) · Ethereum (ETH)

1. The AI Mirage: How Machine Learning Fools Quants

- **The Bogus Claim:** An AI agent pitched a linear model boasting an astounding $R^2 = +47.56\%$ cross-validated edge.
- **The Look-Ahead Catastrophe:** The model used a centered rolling window (`trend5 = r.rolling(5, center=True).mean()`), directly peeking 2 days ahead!
- **The Impact:** Fixing this single line collapsed R^2 from $+47.56\%$ to -0.22% (-47.78 **percentage points** wiped out).
- **5 Other Critical Flaws:** Full-sample scaling, full-sample median splits, shuffled K-fold splits, tuning on holdout, and unanchored benchmark.

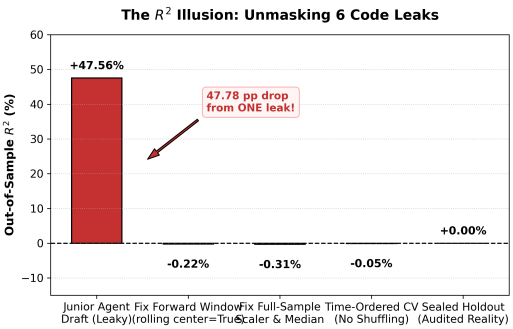


Figure 1: Fixing forward look-ahead wipes out the entire claimed edge.

Executive Governance Lesson

Code that compiles without errors can be financially lethal. High out-of-sample R^2 in daily returns is almost always a data leak, never an alpha discovery.

2. The Feature Factory: Strict As-Of Auditing Architecture

- **50+ Feature Factory:** Engineered return momentum, rolling volatility, volume trends, calendar flags, and cross-asset spillovers.
- **Agent Proposal Audit:** Audited three AI proposals with strict automated truncation tests:
  - `vol_regime_leaky`: Full-sample median $\rightarrow$ **FAILED Audit** ($|diff| > 0$).
  - **The Repair:** 252-day trailing median $\rightarrow$ **PASSED Audit** ($|diff| = 0.000$).
  - `rsi14` & `rel_vs_spy`: **PASSED** clean.
- **Pitfall of Naive t-tests:** Standard t-tests across 5 folds falsely inflate statistical significance due to overlapping sample dependence.

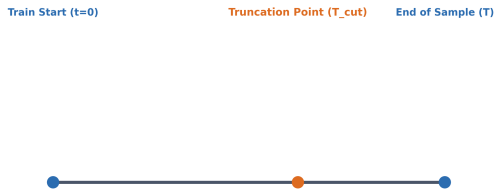


Figure 2: The As-Of Truncation Audit catches leaks before models train.

The As-Of Invariance Axiom

An information set I_t must never change historical values when future observations $t+1, \ldots, T$ are truncated.

3. The Tournament: Why Sound ML Selects Zero Features

- **Time-Ordered CV:** 5-fold expanding window cross-validation (respecting market chronology, zero shuffling).
- **Ridge vs Lasso Showdown:** Penalizing model complexity across $\alpha \in [10^{-4}, 10^{2}]$ to prevent spurious curve-fitting.
- **The 1-Standard-Error Rule:** Choosing the simplest model within 1 SE of minimum CV error (Hastie et al.).
- **The Verdict: Empty Stable Set!** At the 1-SE penalty, Lasso sets **all coefficients to exactly zero** across all 5 assets.
- **First-In vs Stable:** Features that enter first (e.g. lag-1 on ATHEX) represent in-sample noise that fails to generalize across folds.

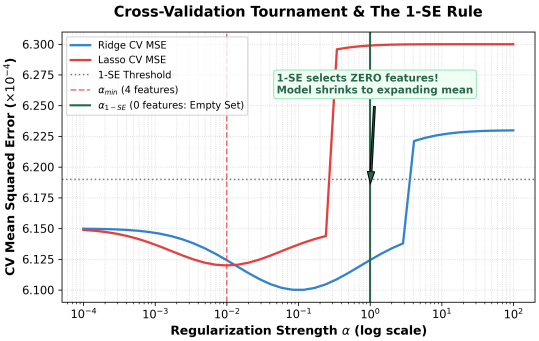


Figure 3: CV tournament curves showing 1-SE shrinkage to zero features.

Occam's Razor in Finance

"When signal-to-noise is near zero, the optimal machine learning model is the expanding historical average."

4. Holdout Showdown: Can Linear ML Beat the Benchmark?

- **Sealed Holdout Protocol:** 2-year forward window (Oct 2024 – Oct 2026, $N = 489$ to 723 days) with quarterly walk-forward refits ($refit = 63$).
- **The Competitors:** Tuned Ridge and Lasso vs. Campbell–Thompson Expanding Historical Mean.
- **The Empirical Results:**
  - **SPY:** Mean RMSE 75.4 bps vs Lasso 75.4 bps ($p = 1.00$).
  - **EUR/USD:** Mean RMSE 44.5 bps vs Lasso 44.5 bps ($p = 1.00$).
  - **BTC & ETH:** Lasso ties or loses slightly ($p > 0.88$).
  - **ATHEX:** No statistically significant edge ($p = 0.69$).
- **Diebold–Mariano Test:** Confirms zero alpha across the entire multi-asset universe ($p \gg 0.05$).

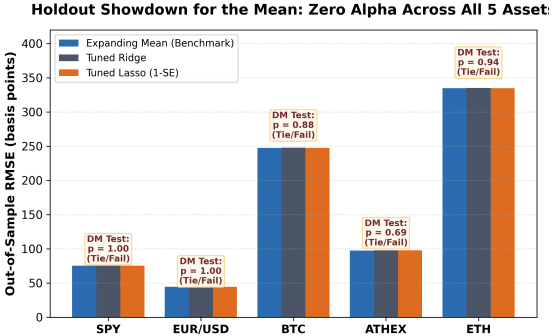


Figure 4: Out-of-sample RMSE and Diebold–Mariano test results.

Quantitative Reality

5. Showdown for Risk: Why Econometrics Crushes Linear ML

- **The Great Financial Asymmetry:** Return direction has zero memory, but **volatility has vast persistence and clustering**.
- **The Risk Tournament:** Testing Constant Vol, Rolling-21, EWMA, Linear Lasso-var, and GARCH(1,1)-t.
- **Robust Evaluation via QLIKE Loss:**
  - QLIKE is scale-invariant and robust to extreme fat tails and jumps (Patton, 2011).
  - Linear Lasso-var on squared returns fits poorly due to heavy noise in squared residuals.
- **The Winner: GARCH(1,1)-t** dominates across all 5 assets, cutting loss by up to 25% over static estimates.

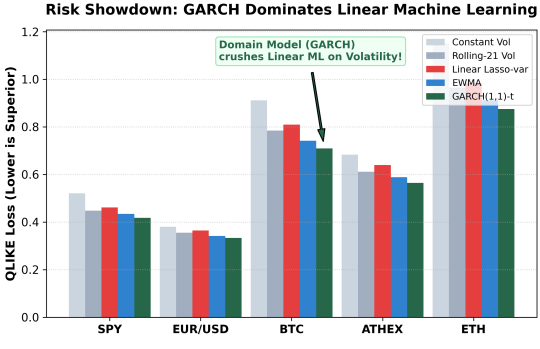


Figure 5: QLIKE risk loss across models: GARCH dominates linear ML.

The Risk Mandate

While ML fails on directional return prediction, structured econometric volatility models provide actionable, high-conviction risk forecasts.

6. Executive Synthesis: The Auditing ML Suite v1.0

- **Behavioral Traps in Quantitative Finance:**
  - *Confirmation Bias:* Quants stop debugging when R^2 is high.
  - *Narrative Fallacy:* Inventing macro stories for random noise.
  - *Garden of Forking Paths:* Trying 100 features guarantees a false discovery at the 5% level.
- **Auditing ML Suite v1.0 (8 Mandates):**
  1. Zero future look-ahead in rolling indicators.
  2. Strict As-Of truncation invariance test.
  3. Preprocessing pipelines sealed inside CV folds.
  4. Time-ordered expanding folds (zero shuffling).
  5. 1-Standard-Error regularization rule.
  6. Campbell–Thompson expanding historical benchmark.
  7. Diebold–Mariano statistical significance testing.
  8. Econometric GARCH for risk management.

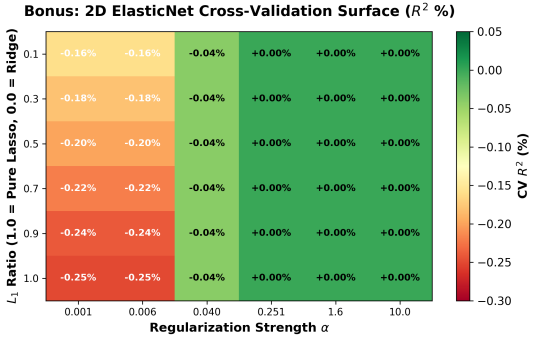


Figure 6: 2D ElasticNet surface confirms noise dominance across all α and L_1 ratios.

Final Investment Conclusion

Deploy historical mean for return budgets; deploy GARCH(1,1)-t for dynamic risk management and capital allocation.